



UN20 Serial™ Fingerprint Management System Developer's Guide

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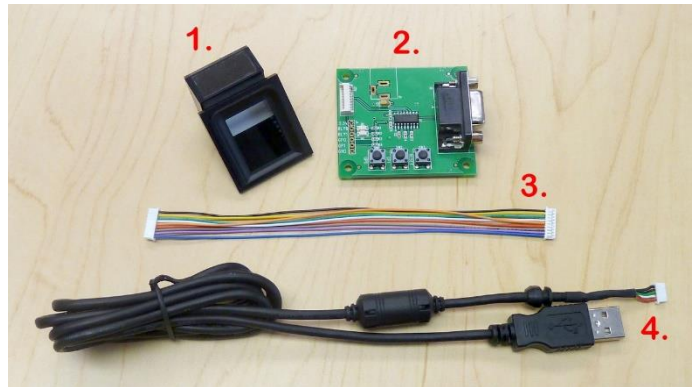
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Chapter 1. Product Overview

The SecuGen® UN20 Serial™ OEM fingerprint module is an advanced stand-alone fingerprint recognition system that captures and stores fingerprint data and identifies or verifies registered users. Designed to function entirely without an external PC, the UN20 Serial includes a fingerprint optic module with a built-in processor board that includes onboard CPU, SDRAM, and flash memory.



UN20 Serial



UN20 Serial Developer Kit hardware

1.1. UN20 Serial Developer Kit Contents

The UN20 Serial Developer Kit consists of hardware and software needed to integrate the UN20 Serial into a wide variety of “embedded fingerprint” applications.

The UN20 Serial Developer Kit for Fingerprint Management System (FMS) runs on Windows 10 / 8.1 / 7 to help programmers quickly and easily build reliable applications based on UN20 Serial’s Fingerprint Management System.

1. **UN20 Serial** with MIPS CPU, RAM and Flash Memory
2. **Development board** with RS232 DB9 connector, LEDs, GPIO buttons and RESET button
3. **9 Pin Molex cable** to UN20 Serial to development board
4. **USB cable** to connect to +5VDC power
5. **UN20 Serial Fingerprint Management System Development Kit**
 - SDATest Demo Program (Windows 10 / 8.1 / 7)
 - Sample fingerprint template database
 - Documentation
 - C# Sample Source Code (Windows 10 / 8.1 / 7)

1.2. Documentation Map

	UN20 Serial DK Fingerprint Management System DK
Introduction Development Options Integration Examples	<i>Quick Start Guide (main folder)</i>
Hardware Specifications Connector Pinouts Power Requirements	<i>UN20 Serial Datasheet (main folder)</i> <i>UN20 DK Board Datasheet (main folder)</i>
Developer's Guide SDK Contents SDATest Guide FMS Protocol Quick Start Guide	<i>This Document</i>
FMS Protocol Manual FMS Packet Structure FMS Commands FMS Error Codes	<i>FMS Protocol Manual</i>

Chapter 2. UN20 Serial PC Demo Programs

2.1. SDATest DK Demo for Windows – Fingerprint Controller

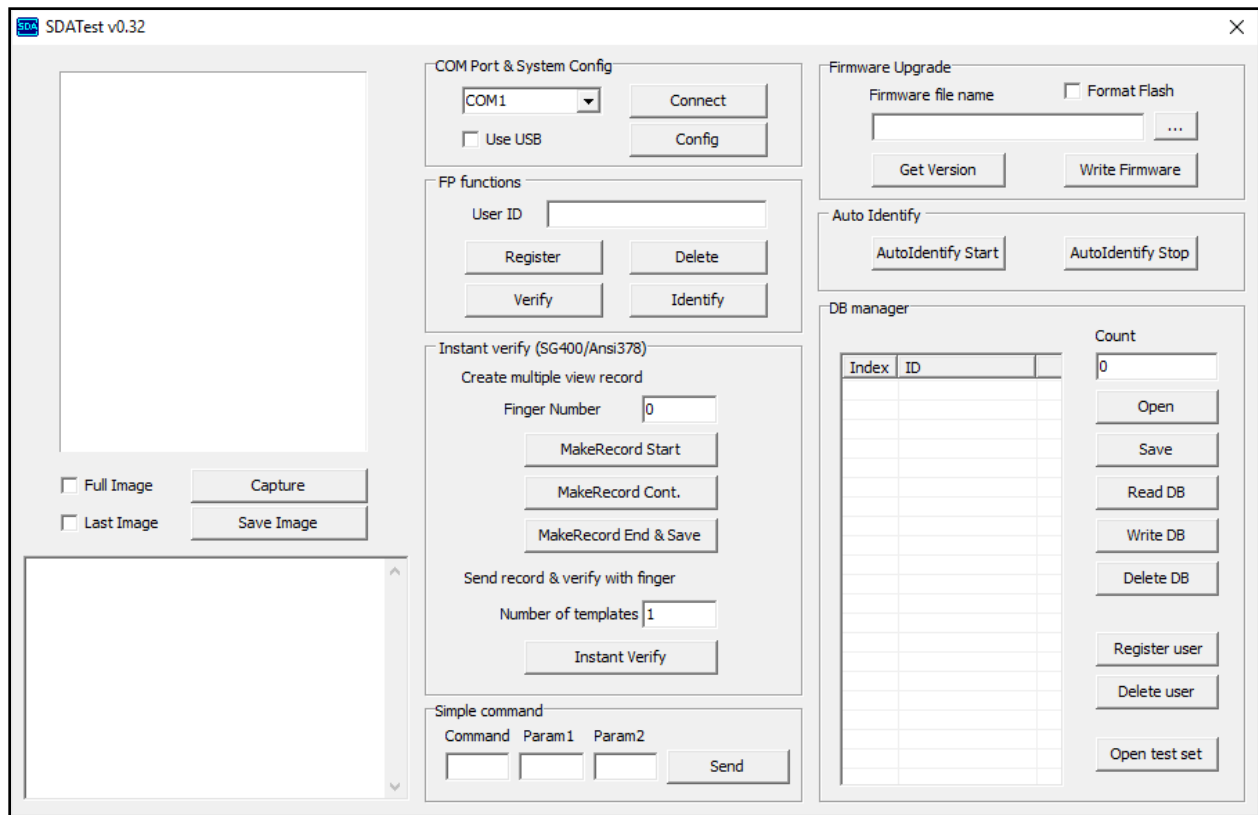
The SDATest Fingerprint Controller is included in the Developer's Kit, and runs on Windows 10 / 8.1 / 7. It uses a sample program to demonstrate the basic fingerprint controls and functions of SecuGen's UN20 Serial OEM module.

To run the SDATest program, do the following:

- Use a computer with Windows 10 / 8.1 / 7 installed.
- Copy SDATEST.EXE from the DK distribution to a directory on your computer
- Double-click SDATEST.EXE.

The SDATest Fingerprint Controller performs the following functions:

- Initializes connection to the UN20 Serial
- Registers, verifies, deletes and updates users
- Sets system configuration

Fingerprint controller (SDATest.exe) User Interface

Note: SDATEST v0.32 or later must be used.
Earlier versions will not work with UN20 Serial.

2.1.1. Initializing a Serial Connection

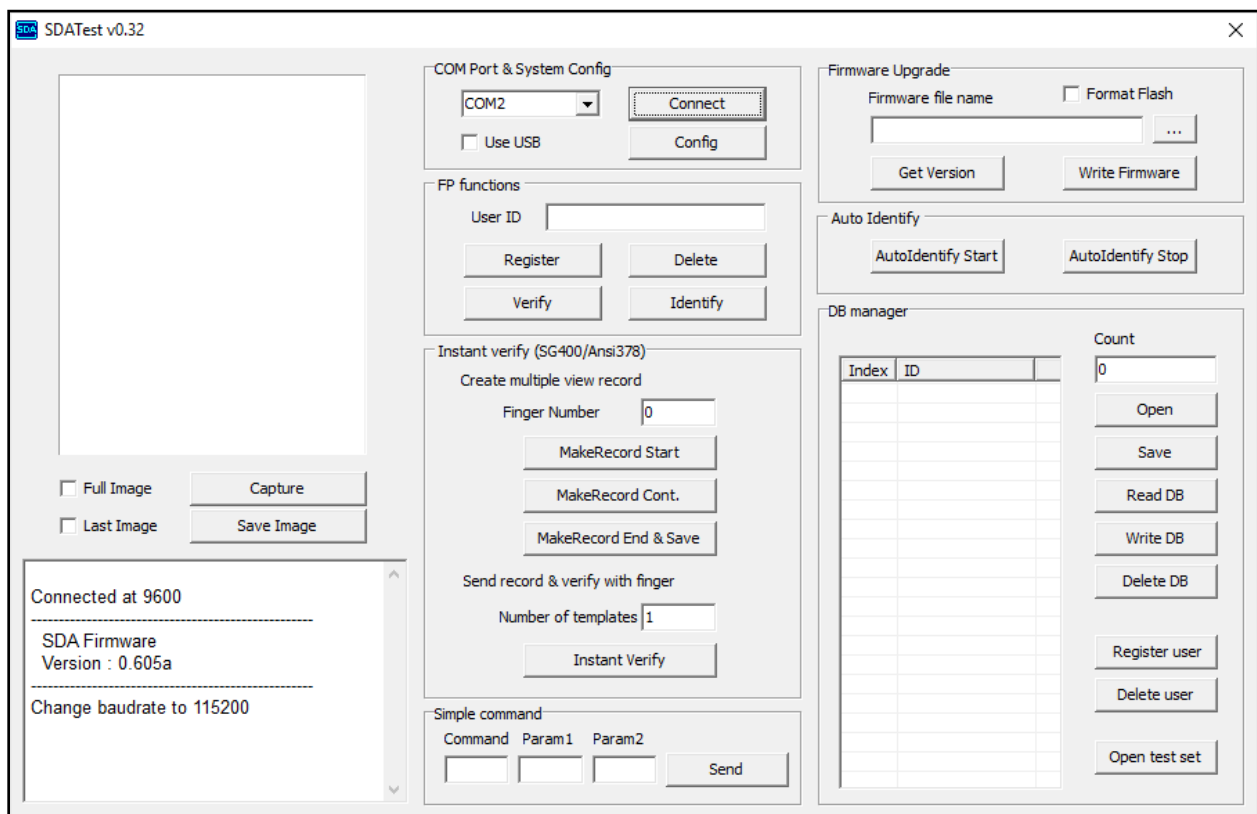
The first step performed by the UN20 Serial Fingerprint Controller is initializing communication between the UN20 Serial and the host computer system. The serial port and communication speed must be selected to start the program. If initialization is successful, the program proceeds to the next step. If initialization is not successful, check the serial port selection and try again. After changing the serial port connection (e.g. from COM1 to COM4), the communication speed will be automatically detected and reset to the appropriate value so there should be no need to select it.

Serial Port Setting – Choose the serial port to which the UN20 Serial is connected (COM1, COM2, COM3 or COM4).

Communication Speed – UN20 Serial can communicate at speeds of 1200 to 921600bps. The default on initial power on is 9600 bps. UN20 Serial communicates at 115200bps

1. Click **Connect**

When connecting to UN20 Serial, you should receive the message “Connected at 9600” followed by the firmware version string. After the initial connection SDATEST.EXE sets the baud rate to 115200bps and reconnects to the module automatically.



2.1.2. System Configuration

Several system configuration settings can be changed by administrators, including brightness, security level, communication speed, master verification, and smart capture. All of these controls are found in the Configuration Dialog window below.

1. Click **Config**

The screenshot shows a window titled "System Info Configuration" with a list of parameters and their values:

SI_USE_MASTER AUTHENTICATION	OFF	SI_USE_RELAY	OFF
SI_COM_SPEED	115200	SI_RELAY_TIME (sec)	0
SI_TEMPLATE_TYPE	SG400	SI_WIEGAND_FORMAT	26 Bit
SI_USE_DATA_CHECKSUM	OFF	SI_WIEGAND_SITE_CODE	0
SI_REGISTER_QUALITY	40	SI_Check_Duplicat_template	
SI_VERIFY_QUALITY	30	SI_Diff_FP_Register	OFF
SI_IDENTIFY_SECU_LEVEL	5	SI_SENSOR_TIMEOUT (sec)	5
SI_VERIFY_SECU_LEVEL	5	SI_ADAPTIVE_CAPTURE	ON
SI_BRIGHTNESS	50	SI_USING_AUTO_ON	ON

At the bottom right, there are "Cancel" and "OK" buttons.

Refer to the **FMS Protocol Manual** for a detailed description of these parameters and how they affect the functionality of the UN20 Serial OEM fingerprint module.

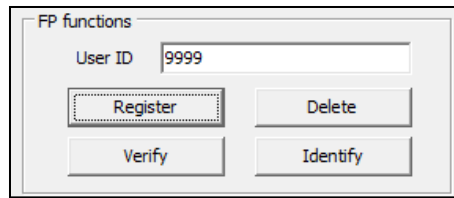
2.1.3. User Registration

User registration, or enrollment, is the process by which new users are added to the FMS database. During registration, two fingerprint images from the same finger are captured and stored in the UN20 Serial flash for future verification and identification operations. The user is required to release the finger from the sensor momentarily before the second (final) capture is performed.

Tips for Using SecuGen Fingerprint Readers

- Cover the fingerprint sensor completely with your finger.
- Don't press too hard. The amount of pressure required is only as much as it takes to hold a piece of paper between your thumb and finger – not much.
- Don't move your finger during scanning.

1. Enter User ID – example 9999



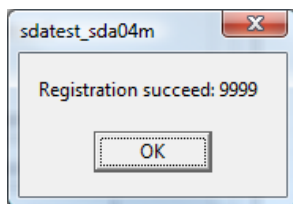
2. Click **Register**



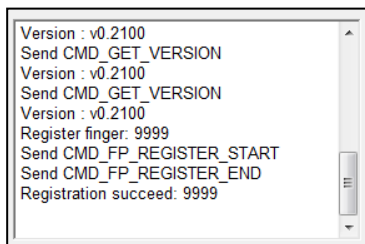
3. Place finger on sensor and then click **OK**. First fingerprint sample is captured.



4. Remove finger and place on sensor again to capture second sample. Click **OK**.



5. Registration confirmation message will be displayed and transaction results will be displayed in the output text box.



Note: Two fingerprints from the same finger must be captured to ensure accuracy and reliability during the registration (enrollment) process.

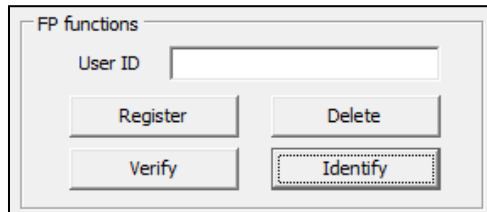
2.1.4. User Verification and Identification

This function compares a captured (input) fingerprint to the fingerprints currently registered in the UN20 Serial database and may perform either of two different operations: identification or verification.

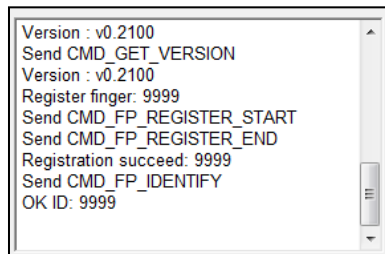
- **Identification** is a one-to-many (1:N) matching operation that does not require a User ID as input. This function compares a captured fingerprint to the registered fingerprints stored in the system to determine the eligibility of a user. Identification works well with relatively small databases. The UN20 Serial uses the SecuGen SecuSearch engine enables extremely fast identification speeds.
- **Verification** is a one-to-one (1:1) matching operation that requires a User ID as input to find a specific user's fingerprint.

Example A. Identification (1:N matching)

1. Place finger on fingerprint sensor
2. Click **Identify**



3. The User ID of the user that was identified will be returned



4. If the user does not exist a "Failure" message will be displayed in the output text box.

```
Version : v0.2100
Register finger: 9999
Send CMD_FP_REGISTER_START
Send CMD_FP_REGISTER_END
Registration succeed: 9999
Send CMD_FP_IDENTIFY
OK ID: 9999
Send CMD_FP_IDENTIFY
Fail.
```

Example B. Verification (1:1 matching)

1. Enter the User ID.

FP functions

User ID: 9999

Register Delete

Verify Identify

2. Place finger on sensor
3. Click **Verify**. The fingerprint will be captured and matched against the corresponding record in the database. The match result will be output to the text box.

```
Send CMD_FP_REGISTER_START
Send CMD_FP_REGISTER_END
Registration succeed: 9999
Send CMD_FP_IDENTIFY
OK ID: 9999
Send CMD_FP_IDENTIFY
Fail.
Send CMD_FP_VERIFY
Succeed Verify finger: 9999
```

2.1.5. User Deletion

This function is used to delete previously registered users.

1. Enter the User ID.
2. Click **Delete**.

3. The user record will be deleted and the result output to the text box.

4. To delete all registered users from the user database at once, click Delete DB. In this case you do not need to specify a User ID, but a precautionary confirmation dialog will appear before all users are deleted.

2.1.6. Generate Multi-View ANSI378 Template

1. Click **Config**
2. Set SI_TEMPLATE_TYPE to ANSI378

SI_TEMPLATE_TYPE ANSI378

3. Click **OK**
4. Click **MakeRecord Start**

SDA Test v0.22

COM Port & System Config: COM4, Connect, Config

Firmware Upgrade: Firmware file name, Format Flash, Get Version, Write Firmware

Auto Identify: AutoIdentify Start, AutoIdentify Stop

DB manager: Index, ID, Count (0), Open, Save, Read DB, Write DB, Delete DB, Register user, Delete user, Open test set

Instant verify (SG400/Ansi378): Create multiple view record, Finger Number (0), MakeRecord Start, MakeRecord Cont., MakeRecord End & Save, Send record & verify with finger, Number of templates (1), Instant Verify

Simple command: Command, Param1, Param2, Send

Full Image, Capture, Last Image, Save Image

Connected at 115200
SDA Firmware Version : 0.2100
Send CMD_MAKE_RECORD_START OK

5. Enter finger number for finger view 1
6. Place finger1 on sensor. Fingerprint sensor will capture sample of finger 1
7. Click **MakeRecord Continue**

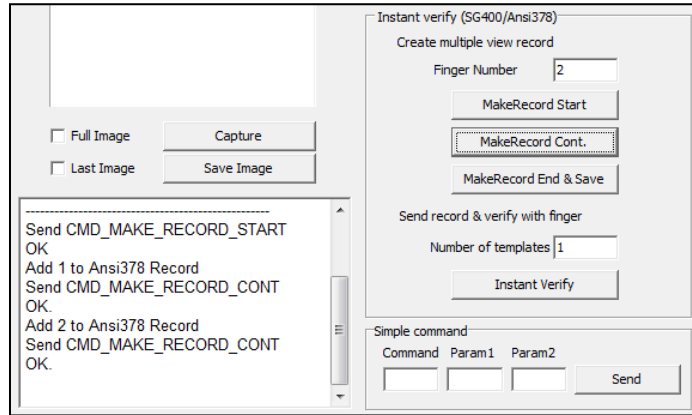
Instant verify (SG400/Ansi378): Create multiple view record, Finger Number (1), MakeRecord Start, MakeRecord Cont., MakeRecord End & Save, Send record & verify with finger, Number of templates (1), Instant Verify

Simple command: Command, Param1, Param2, Send

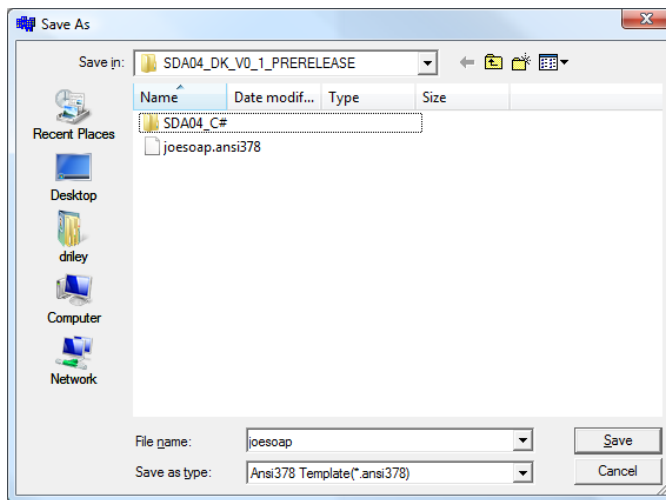
Full Image, Capture, Last Image, Save Image

SDA Firmware Version : 0.2100
Send CMD_MAKE_RECORD_START OK
Add 1 to Ansi378 Record
Send CMD_MAKE_RECORD_CONT OK

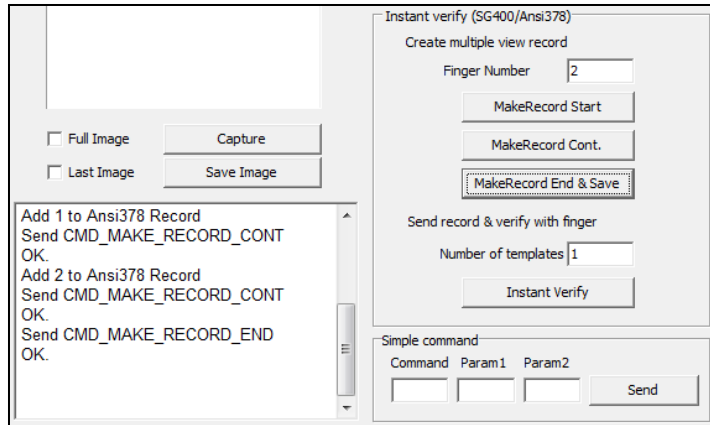
8. Enter finger number for finger view 2
9. Place finger2 on sensor
10. Click **MakeRecord Continue**



11. Click **MakeRecord End & Save**

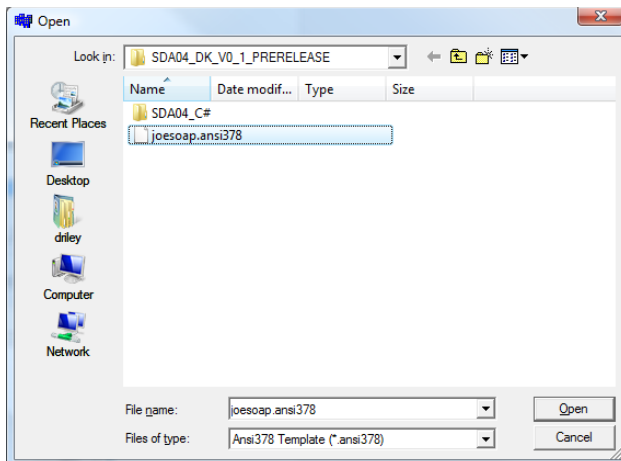


12. Enter file name and click **Save** to write record to disk. OK status should be displayed.

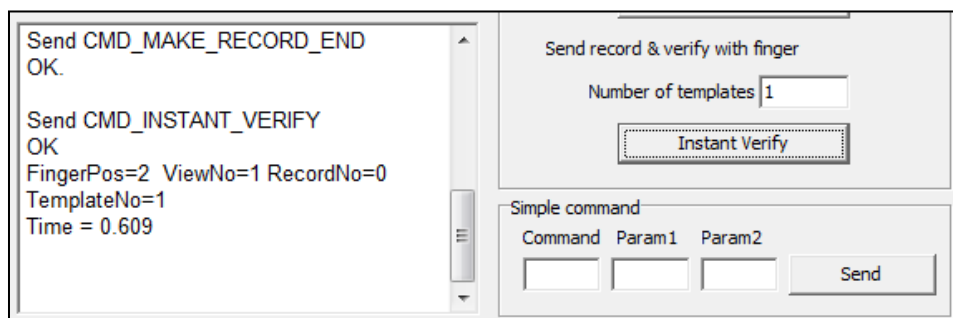


2.1.7. Verify ANSI378 Template

1. Place finger on sensor
2. Click **Instant Verify**
3. At the **Open** dialog, select ANSI template and click **Open**



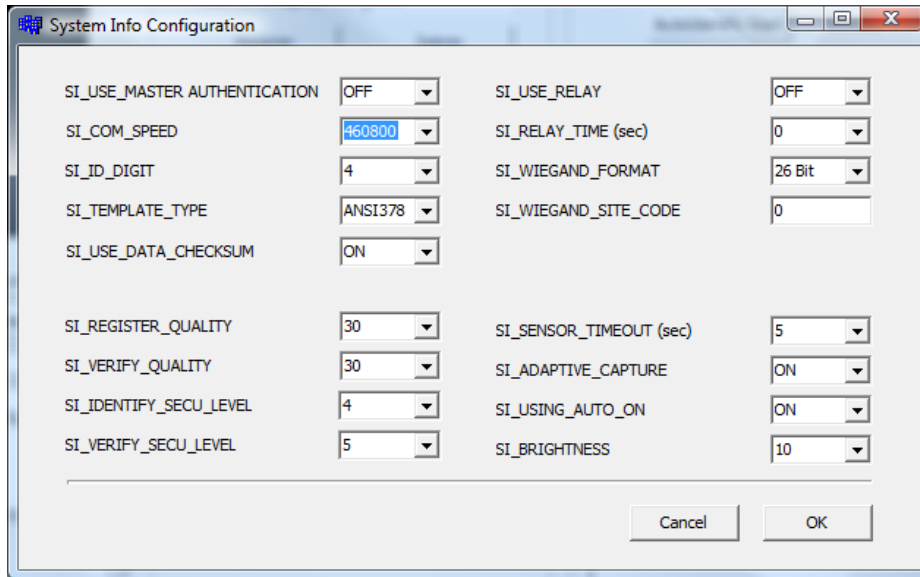
4. Live fingerprint will be extracted and matched against the ANSI record. Finger position and view will be reported in the result.



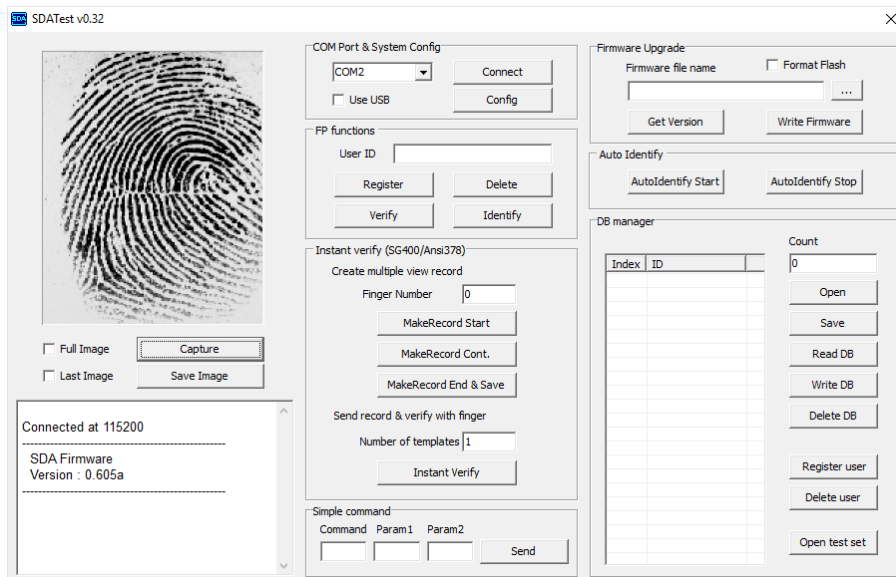
2.1.8. Capturing and Storing Fingerprints

When capturing a full size image, it is preferable to use high speed serial communications whenever possible to ensure fastest possible image transfer.

1. Click **Config** and set baud rate to 460800bps, then click **OK**



2. Select **Full Image**
3. Place finger on sensor
4. Click **Capture**

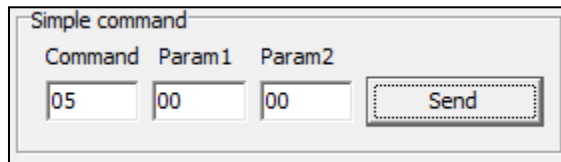


5. If you wish to store the fingerprint image, click **Save Image** to save the image as a TIF file.

2.1.9. Sending Simple Commands

When capturing a full size image, it is preferable to use high speed serial communications whenever possible to ensure fastest possible image transfer.

1. Connect to UN20 Serial using SDATEST.EXE.
2. Enter a command in the Simple Command controls. The example below sends CMD_GET_VERSION

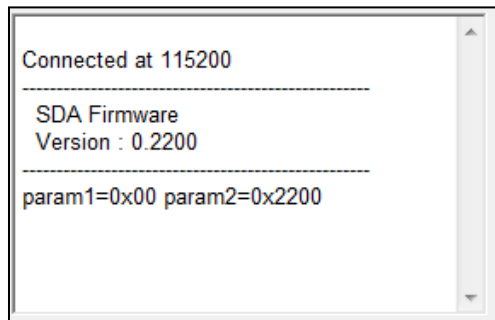


Simple command

Command	Param1	Param2
05	00	00

Send

3. Click **Send**. Results will be displayed in the text control.



Connected at 115200

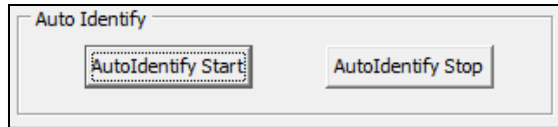
SDA Firmware
Version : 0.2200

param1=0x00 param2=0x2200

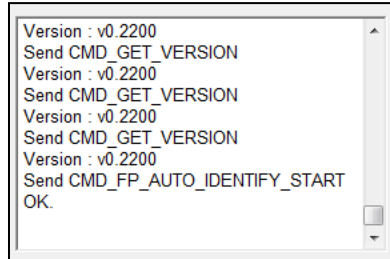
2.1.10. Auto Identifying

In Auto Identification mode, whenever a finger is placed on the sensor, the fingerprint is captured and matched against the database. If a match is found, the user ID is returned to the host. If a match is not found, an error is returned. After the result is returned to the host, the system waits for the next fingerprint until Auto Identification is disabled.

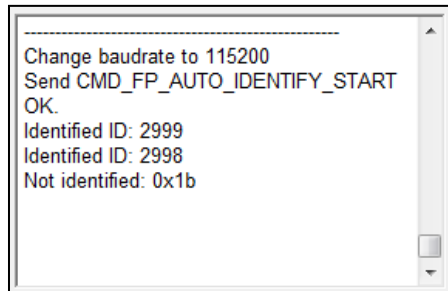
1. Connect to UN20 Serial using SDATEST.EXE.
2. Click **Autoidentify Start**



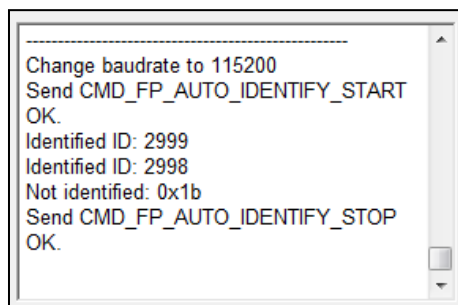
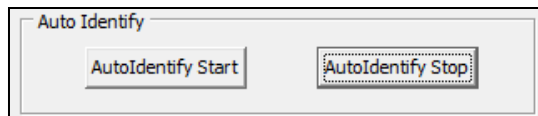
3. UN20 Serial will enter Auto Identification mode



4. Place a finger on the sensor. Failed identifications will return 0x1b. Successful identifications will return the user ID.



5. Click **Autoidentification Stop** to exit Auto Identification mode.



2.2. SecuSearch 1:N Sample Database

The UN20 Serial DK includes a sample database containing 3,000 users with a total of 6,000 fingerprint templates. This database is provided for the purpose of testing the high speed SecuSearch® engine.

Note: This section is for testing purposes only. All fingerprint data on the UN20 Serial will be erased during the database load process.

2.3.1. Loading the test database

1. Power on UN20 Serial.
2. Launch SDATest.exe and connect to the device.
3. Set Comm. Speed to 460800bps



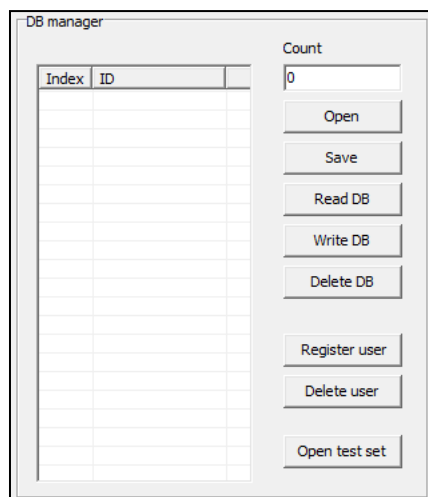
The image shows a control element labeled "SI_COM_SPEED" with a dropdown menu set to "460800".

4. Set Template type to SG400



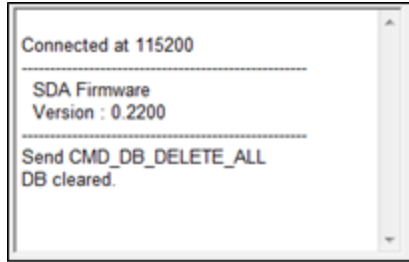
The image shows a control element labeled "SI_TEMPLATE_TYPE" with a dropdown menu set to "SG400".

5. Click **Delete DB** to erase all fingerprint data on the device.

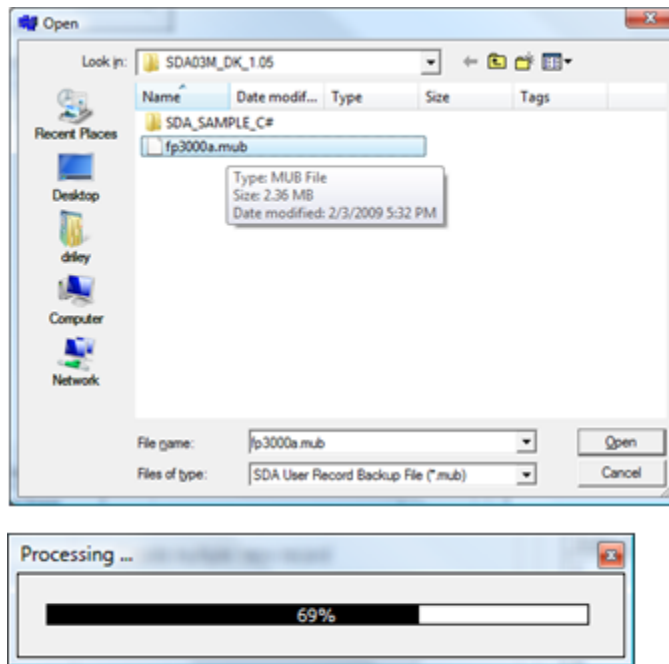


The image shows the "DB manager" window. It contains a table with columns "Index" and "ID". To the right of the table is a "Count" field showing "0". Below these are several buttons: "Open", "Save", "Read DB", "Write DB", "Delete DB", "Register user", "Delete user", and "Open test set". The "Delete DB" button is highlighted.

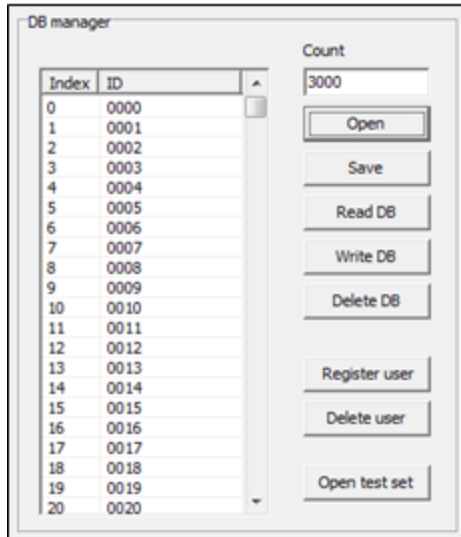
6. Confirm that database was successfully cleared.



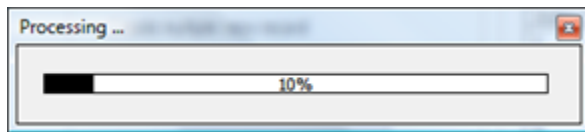
7. Click **Open** to select the test database. Open fp3000a.mub.



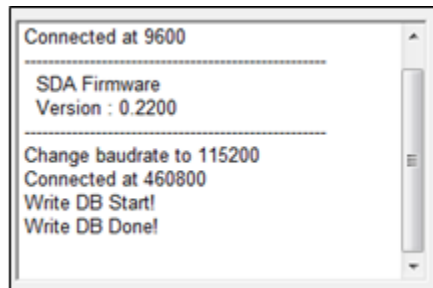
8. Fingerprint templates will be loaded into SDATest.EXE memory in preparation for writing to the UN20 Serial.



9. Click **Write DB** to write the template database to the UN20 Serial.

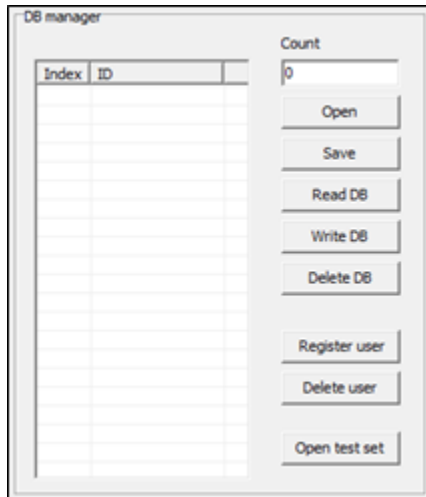


10. After successful write operation, there should be no errors reported in the text box.

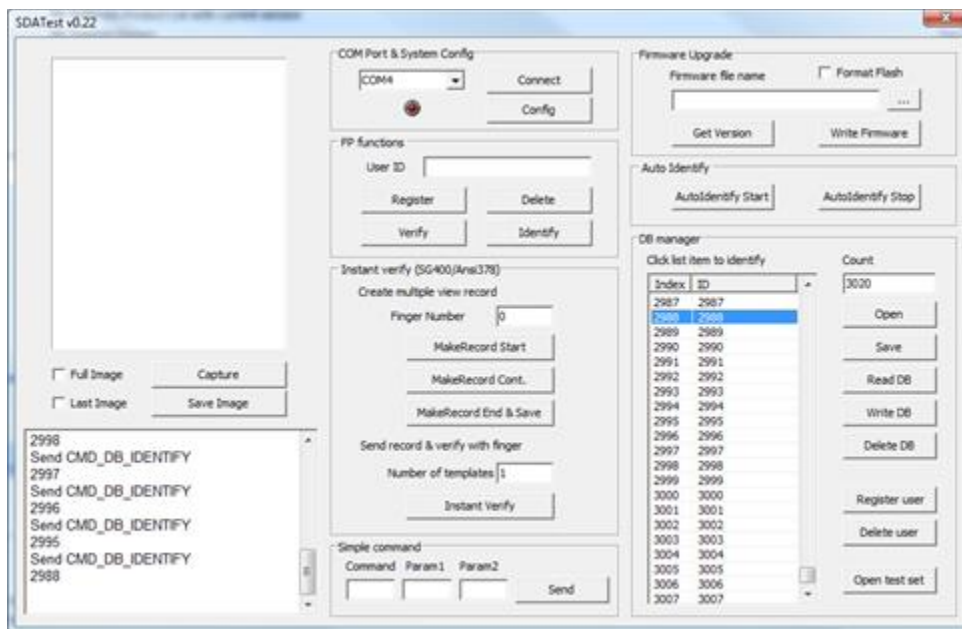


2.3.2. Identifying fingerprints in the test database

1. Click **Open test set** and load the test database into SDATEST memory. FP3000.tdb for UN20 Serial.



2. Double-Click on any template in the grid control to send it to the UN20 Serial for identification. Identification results will be displayed in the text control.



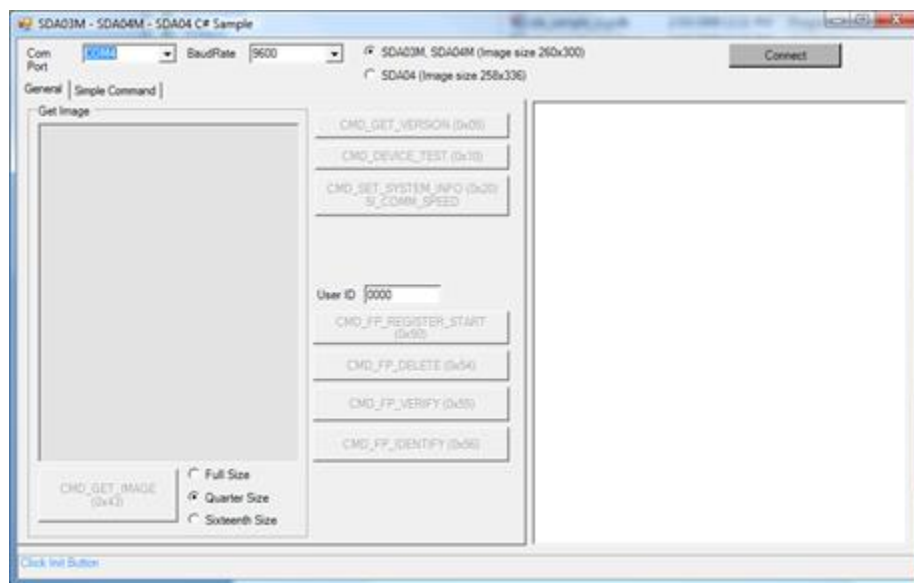
3. Since the test database is fully loaded, it is advisable to delete the last template loaded and register your own fingerprint to test identification with the fingerprint capture. Click Register and follow the registration process as described in [Section 2.1.3](#) above

2.4 C# Sample Application including source code

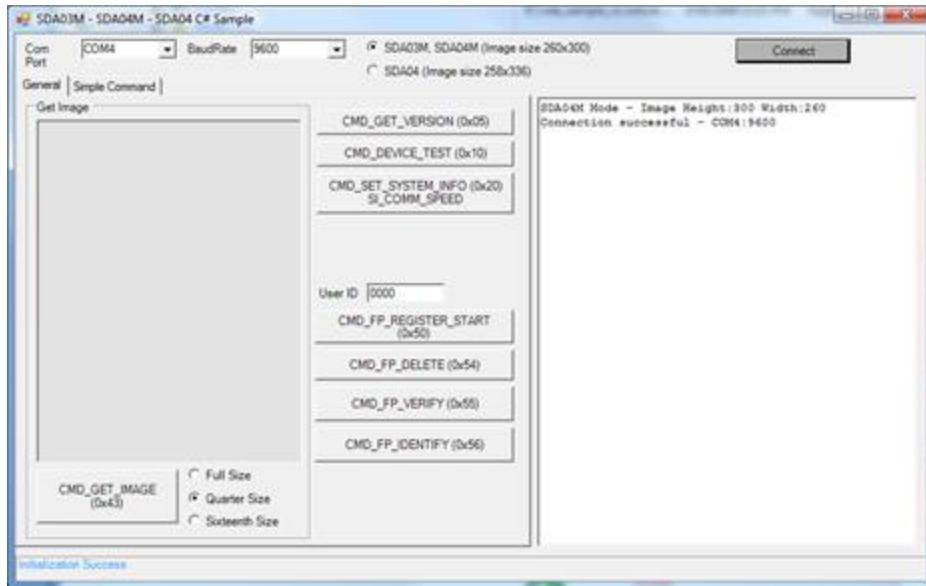
The UN20 Serial DK provides sample source code for a host controller implemented in C#. This application can be built on Windows 10 / 8.1 / 7 using Microsoft Visual Studio 2012 or later. The C# sample is not as full-featured as SDATEST.EXE but it contains enough functionality to allow a developer to view code and port to the target host controller.

2.4.1. Initializing Connection

1. Connect the UN20 Serial device to a serial port on your PC and power it on.
2. Select the following:
 - a. COM port that the UN20 Serial is connected to
 - b. 9600 baud rate
3. Click **Connect**



4. After successful connection, the buttons in the sample application will be enabled.



5. Click ***CMD_GET_VERSION*** to get the FMS firmware version.

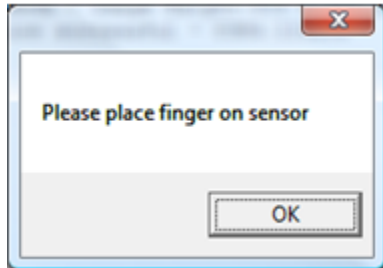
```
SDA04M Mode - Image Height:300 Width:260
Connection successful - COM4:9600
+++
Send CMD_GET_VERSION
Packet : 00,05,00,00,00,00,00,05
Ack : 00,05,00,2200,00,00,00,27
Total Time: 140ms
Error Code: 00
Version : 00.2200
```

6. Select baud rate of 115200 and then click ***CMD_SET_SYSTEM_INFO*** to set the comm. speed and reconnect to the UN20 Serial.

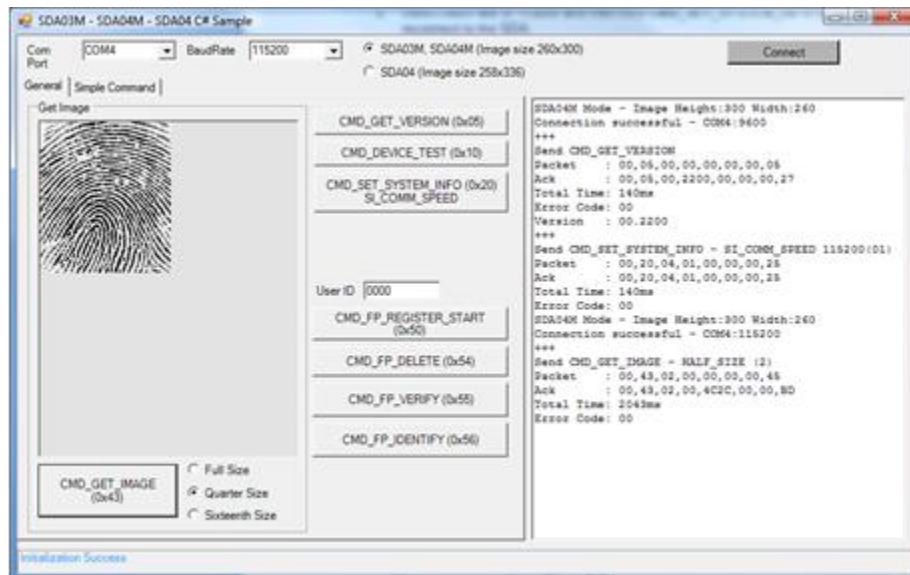
```
SDA04M Mode - Image Height:300 Width:260
Connection successful - COM4:9600
+++
Send CMD_GET_VERSION
Packet : 00,05,00,00,00,00,00,05
Ack : 00,05,00,2200,00,00,00,27
Total Time: 140ms
Error Code: 00
Version : 00.2200
+++
Send CMD_SET_SYSTEM_INFO - SI_COMM_SPEED 115200 (01)
Packet : 00,20,04,01,00,00,00,25
Ack : 00,20,04,01,00,00,00,25
Total Time: 140ms
Error Code: 00
SDA04M Mode - Image Height:300 Width:260
Connection successful - COM4:115200
```

2.4.2. Capturing a fingerprint image

1. Click **CMD_GET_IMAGE**.
2. Place finger on sensor when prompted

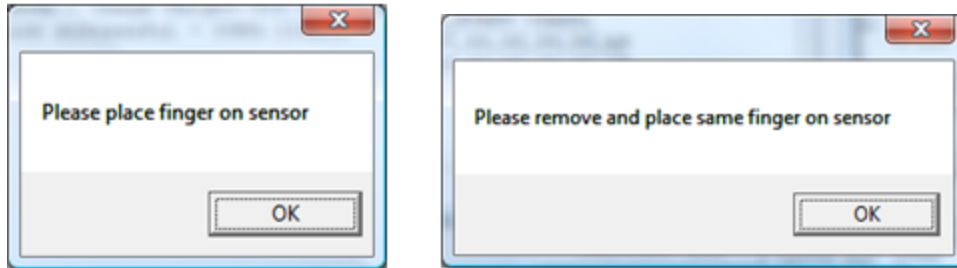


3. The image will be captured and transferred to the host PC. The size of the image depends on the selection of full size or quarter size.

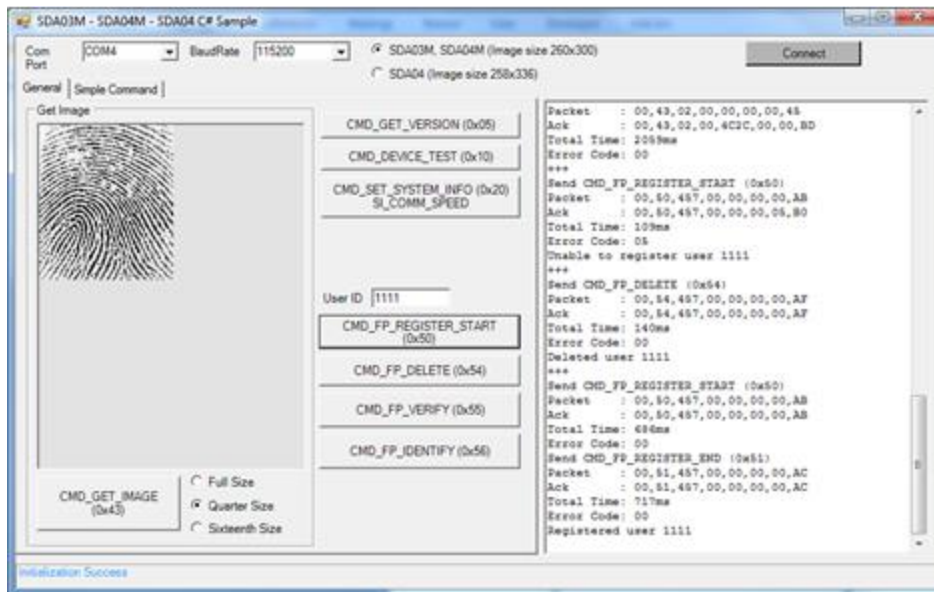


2.4.3. Registering a user

1. Enter a user ID.
2. Click **CMD_FP_REGISTER_START**
3. Place finger when prompted and click **OK**. Repeat finger placement when prompted.



4. Message will be displayed confirming successful registration.



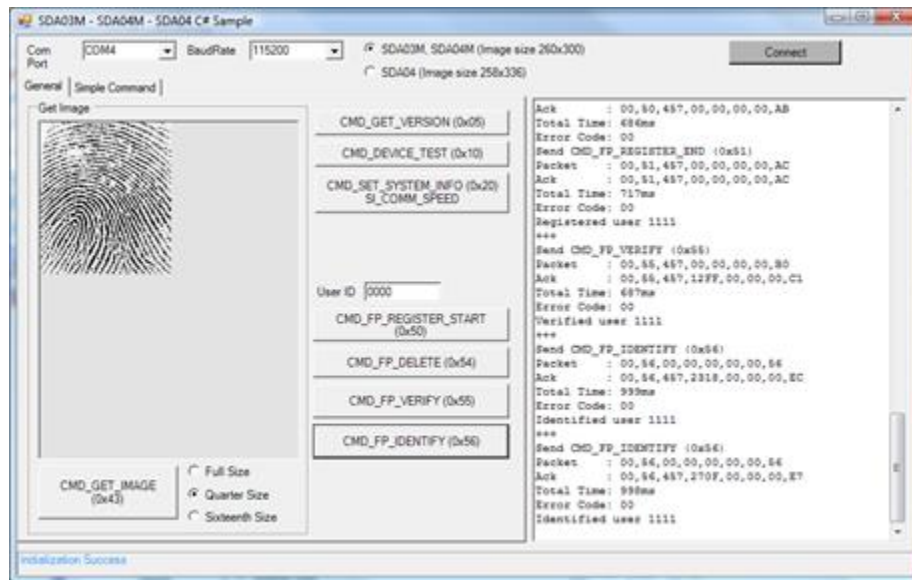
2.4.4. Verifying a user

1. Enter user ID.
2. Click **CMD_FP_VERIFY**.
3. Place finger on sensor when prompted.
4. Verification result will be displayed in the text control.

```
+++
Send CMD_FP_VERIFY (0x55)
Packet : 00,55,457,00,00,00,00,B0
Ack : 00,55,457,12FF,00,00,00,C1
Total Time: 687ms
Error Code: 00
Verified user 1111
```

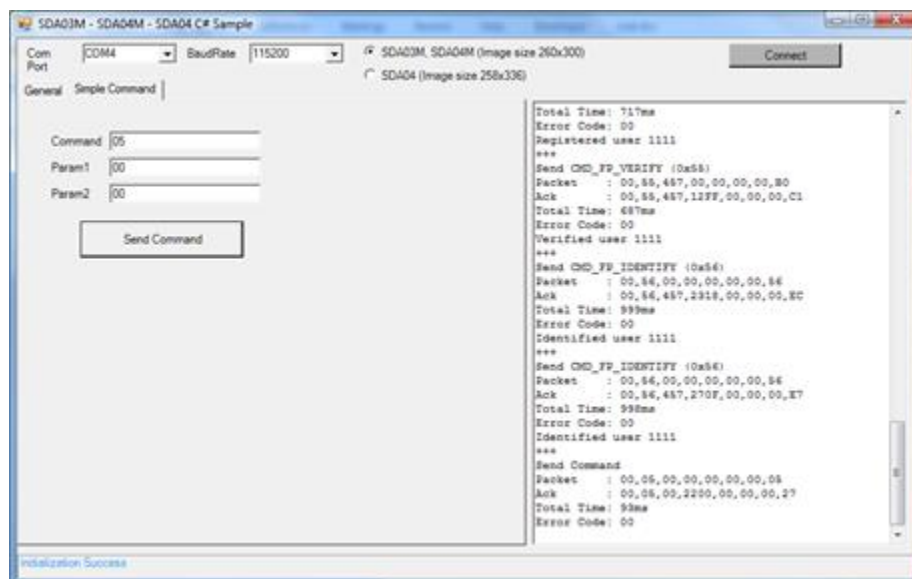
2.4.5. Identifying a user

1. Click **CMD_FP_IDENTIFY**.
2. Place finger on sensor when prompted.
3. Identification result will be displayed in the text control.



2.4.6. Sending Simple Commands

1. Click the **Simple Command** tab.
2. Enter the Command and Param1 and Param2 values.
3. Click **Send Command**.
4. The results of the command will be displayed in the text control.



Chapter 3. Quick Start Guide

Communication between the built-in CPU and the main controller is performed through the serial port. The communication speed can be set to 1200, 2400, 9600, 19200, 38400, 57600, 115200, 230400 or 921600 bps. The default baud rate after initial power on is 9600bps.

All commands are transmitted from the main controller to the UN20 Serial, which then sends acknowledgements for each command back to the main controller using the same command packet format. For the ACK packet, the UN20 Serial transmits the result of the command as an error code in the ErrorCode field. If the command has been executed properly, the ErrorCode value will be 0 (M2ERROR_NONE).

Note: For a complete list of all available commands and descriptions of each, please refer to the **FMS Protocol Manual**.

3.1. Packet Structure

Communication between the UN20 Serial Processing Unit and the main controller is performed using P1 and P2 type packets. The size of the fields (in bytes) and their brief descriptions are listed below. In order to exchange data over 4 bytes, the lwExtraData, hwExtraData fields should be used. The UN20 Serial main controller should receive data of the length specified in the ExtraData field since the value in the ExtraData field indicates this. (For more detailed descriptions of each command, refer to the **FMS Protocol Manual**.)

3.1.1. P1 Packet

1 Byte	1 Byte	2 Bytes	2 Bytes	2 Bytes	2 Bytes	1 Byte	1 Byte
Channel	Command	Param1	Param2	lwExtraData	hwExtraData	ErrorCode	Checksum

3.1.2. P2 Packet (P1 with Extra data)

P1 Packet	data	CS1
-----------	------	-----

Data length is defined by lwExtraData and hwExtraData in the P1 packet area. Normally, extra checksum is disabled by default. Only after enabling this option is the extra checksum added and sent to the host

controller. The host controller also has to send an additional checksum byte when it enables the extra checksum function in the UN20 Serial. Use the option `SI_USE_DATA_CHECKSUM` (0x10) in the command `CMD_SET_SYSTEM_INFO` (0x20).

The values of the first 11 bytes of the packet (excluding the checksum byte) are summed and then divided by 0x0100 (256). This will create a 1-byte shift. The remaining value from this shift is the checksum byte. Checksum can be used to assure correct data transfer for remote registration.

Channel	Always zero
Command	Command field
Param1	The first parameter where data is to be transmitted (if any)
Param2	The second parameter where data is to be transmitted (if any)
lwExtraData	When more than 4 bytes of data are transmitted, this is the data length low word value
hwExtraData	When more data is transmitted, this is the data length high word value
ErrorCode	The executed result of the command
Checksum	Used for communication error detection

In the **Command** field, the command that the processing unit executes is specified. Some commands require a parameter to transmit data, and in this case, this is done through Param1 and Param2. If more data is transmitted in addition to Param1 and Param2 (more than 4 bytes of data), the **lwExtraData** and **hwExtraData** fields are used. The low word value of the length of data is stored in **lwExtraData** field, and the high word value in **hwExtraData**. The **Checksum** field is used to detect communication errors. The values of the first 11 bytes (excluding the checksum byte) are summed and then divided by 0x0100 (256). This will create a 1-byte shift. The remaining value from this shift is the checksum byte.

3.2. Packet Transfer

All commands are transmitted to the UN20 Serial from the main controller. The UN20 Serial replies to each command with an acknowledgement using the same format as the command packet. (The command packet is sent to the UN20 Serial from the main controller, and the ACK packet is sent to the main controller from the UN20 Serial.) The UN20 Serial sends the result of the command in the **ErrorCode** field of the ACK. If execution of the command is completed without error, the **ErrorCode** is '0' (`M2ERROR_NONE`).

Example

The following representation shows the command packet when using the command **CMD_FP_VERIFY (0x55)** to verify a user whose User ID is 1234. The command is 0x55, and Param1 is 0x1234.

Channel	Command	Param1					
0x00	0x55	0x1234	0x0000	0x0000	0x0000	0x00	Chksum

As mentioned previously, during word transmission the high word value is transmitted followed by the low word value. Therefore, the byte sequence of actual transmission is as follows:

0x00 → 0x55 → **0x34** → **0x12** → 0x00 → 0x00 → 0x00 → 0x00 → 0x00 → 0x00 → 0x00

In most cases, one command packet executes one function. But in some cases several command packets are needed to complete a function. Commands such as Master Register, User Register, Change of User Fingerprint, User Delete and System Mode Change, among others, require master authentication first and can only be executed independently if master authentication is disabled by CMD_SET_SYSTEM_INFO.

3.3. Commands

Commands are classified into one of the following four categories depending on the parameter.

- Without parameters
- Transmitting values to FMS
- Obtaining values from FMS
- Exchanging data over 4 bytes in size

► *Without parameters*

CMD_GET_VERSION

This command is used to check the current version of the FMS.

Command Packet from the Main Controller

0x00	0x05	0x0000	0x0000	0x0000	0x0000	0x00	Chksum
------	------	--------	--------	--------	--------	------	--------

The UN20 Serial receives and executes the CMD_GET_VERSION command and then sends out CMD_GET_VERSION in the command field and the execution results in the ErrorCode field all in the same packet format. If the command has been properly executed, the ACK packet from the FMS to the main controller will be as follows:

ACK Packet of FMS (Major, Minor Version Number in Param1 & Param2)

0x00	0x05	0x0000	0x100A	0x0000	0x0000	0x00	Chksum
------	------	--------	--------	--------	--------	------	--------

► **With parameters**

CMD_RELAY_ONOFF

This command controls use of the relay function. The main controller sends out the relay number to Param1 of the command packet, and determines whether or not to send the relay function to Param2, then to the UN20 Serial. The UN20 Serial executes the command after receiving this command packet, and then returns the execution result in the ErrorCode field of the ACK Packet.

The example below opens relay No. 1 (0x0001).

Command Packet of the Main Controller

0x00	0x12	0x0001	0x0001	0x0000	0x0000	0x00	Chksum
------	-------------	---------------	---------------	--------	--------	------	--------

ACK Packet of FMS (Relay No. and Status in Param1 & Param2)

0x00	0x12	0x0001	0x0001	0x0000	0x0000	0x00	Chksum
------	-------------	--------	--------	--------	--------	-------------	--------

► **Getting parameters**

When getting parameters, the FMS sends information requested from the main controller into Param1 or Param2.

CMD_GET_SYSTEM_INFO

This command collects current FMS system configuration values. The example below retrieves the security level.

After being given CMD_GET_SYSTEM_INFO (0x30) in the command field of the command packet, and SI_VERIFY_SECU_LEVEL in the Param1 field, the FMS shows the current security level in Param2 of the ACK packet. The security level is now set at Normal (0x05).

Command Packet of the Main Controller

0x00	0x30	0x0002	0x0000	0x0000	0x0000	0x00	Chksum
------	-------------	---------------	--------	--------	--------	------	--------

ACK Packet of FMS (Security Level Return in Param2)

0x00	0x30	0x0002	0x0005	0x0000	0x0000	0x00	Chksum
------	-------------	---------------	---------------	--------	--------	------	--------

► Exchanging data over 4 bytes in size

In order to exchange data over 4 bytes, the lwExtraData, hwExtraData fields should be used. The FMS main controller should receive data of the length whose value is specified in the ExtraData field.

CMD_GET_IMAGE

This command retrieves the captured FMS fingerprint image data. The main controller determines the size of the image to be acquired (Param1). The possible values are VIEW_NORMAL (0x01, the original image size) and VIEW_QUARTER (0x02, one-fourth of the original size). The size of the data depends on the size of the image chosen. It reads the length of data from the ExtraData field of the packet received from FMS then receives the correspondingly sized image.

Command Packet of the Main Controller

0x00	0x43	0x0001	0x0000	0x0000	0x0000	0x00	Chksum
------	------	---------------	--------	--------	--------	------	--------

ACK Packet of UN20 Serial (Original Image size of UN20 Serial is 0x1d4c0 = 120,000 = 300 x 400)

0x00	0x43	0x0001	0x0000	0xd4c0	0x0001	0x00	Chksum
------	------	--------	--------	---------------	---------------	------	--------

After sending the ACK packet above, the FMS transmits the image data back to the main controller.

Note: For a complete list of all available commands and descriptions of each, please refer to the **FMS Protocol Manual**.

Chapter 4. Best Practices

4.1. Security Levels

Before deploying SecuGen fingerprint recognition devices, administrators should understand the basic concepts behind security levels and their effects on the performance of the fingerprint recognition functions. There will always be a trade-off between user convenience and security.

For example, when you increase the security level, you may experience a higher rate of false rejections (FRR). False rejection occurs when an authorized (registered) user's fingerprint is not successfully matched against the stored sample because more minutiae data (fingerprint characteristics) are required for a match. The increased security level raises the matching threshold, which results in a more discriminating fingerprint recognition system.

On the other hand, when you decrease the security level, you may experience a higher rate of false acceptance (FAR). False acceptance is potentially more serious because this means an unauthorized user is granted access.

The consequence of a "slightly higher security level" could be a slight increase in FRR, which is usually nothing more than an inconvenience to users who should re-attempt to match their fingerprints. But the consequence of a "much higher security level" could be a great increase in FRR, which could pose a hindrance to the smooth operation of a fingerprint recognition system. In high security environments where the security level is raised, users will need to be more attentive to physical technique when placing their fingers on the sensor, and they may require basic education regarding environmental considerations such as bright light and moisture and their possible effects on fingerprint capturing. In summary, where FRR can be irritating, the net results of FAR can be far more serious if critical resources are at stake.

SecuGen devices use security levels to fine-tune the fingerprint matching process, allowing different installations to emphasize security over convenience, or vice versa, depending on the requirements of a site. Security levels range from 1 (Lowest) to 9 (Highest). The default setting of SecuGen devices depends on whether the system is set to use identification or verification as the mode of authentication. For example, the security level in SecuGen's "Identify" mode may range from 6-9, with a default of 8. The default matching mode in SecuGen devices is "Verify," which uses input fingerprint minutiae plus some other data, such as User ID, PIN, or smart card, to quickly compare live fingerprint data (1:1 match) with the enrolled/stored data corresponding to the User ID, PIN or smart card, effectively proving the user's proclaimed identity before granting access. The default security level for verification is 5, or "Normal." Lower security settings may be appropriate for applications with many users, such as time-and-

attendance systems at large-scale sites where high security is not considered as critical as speed and convenience.

Note: FAR and FRR are directly and inversely proportional, so it is important to establish security policies based on a full understanding of this fact.

4.2. ANSI-INCITS 378-2004

The FMS firmware supports the ANSI-INCITS 378-2004 standard finger minutiae format (“ANSI378”). For more information about this fingerprint template standard, refer to the standard document titled “Information technology – Finger Minutiae Format for Data Interchange”, document # ANSI-INCITS 378-2004, available at the ANSI website <http://webstore.ansi.org>.

Chapter 5. Connecting to External Devices

The UN20 Serial supports CMOS level serial, relay, Wiegand and general purpose output connectivity using the 9 pin Molex connector labeled J3.

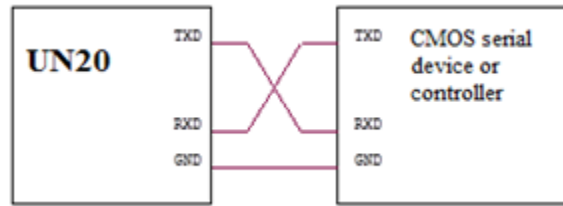
5.1 Connector Description: J3 (9 Pin)

J3 (9 Pin) Connector Description – CMOS Level Serial I/O and GPIO			
Pin No.	Name	States	Description
1	GND	Power	Main Board Power Ground
2	Relay 0	OUT	Relay 0 / Wiegand 0
3	RX	IN	Serial Receive (CMOS Level)
4	TX	OUT	Serial Transmit (CMOS Level)
5	RESET	IN	System Reset Signal (Active Low)
6	VCC	Power IN/OUT	IN: Main Board Power [3.3V DC Regulated] when USB cable is not connected to J4. OUT: Connected Device Power [3.3v DC Regulated] when USB cable is connected to J4.
7	GPIO1	IN/OUT	Programmable GPIO1
8	GPIO0	IN/OUT	Programmable GPIO0
9	Relay 1	OUT	Relay 1 / Wiegand 1

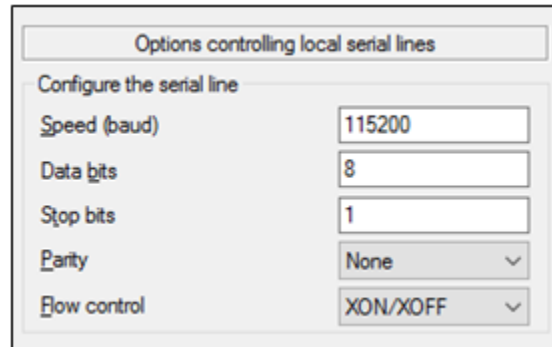
Important:

Although UN20 Serial can be powered using either 3.3VDC on pin 6 of J3 or 5VDC with the USB cable, **it is NOT RECOMMENDED to supply power over both connectors at the same time** because this may cause the device to enter an unknown state. The UN20 Serial supplies 3.3V DC on pin 6 when the USB cable is connected to UN20 Serial.

5.2. UART



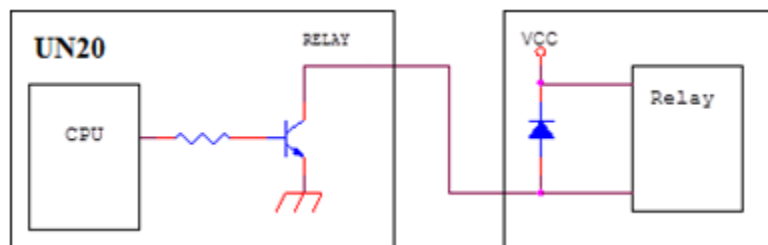
The default communications for UN20 Serial are 115200bps, 8 data bits, no parity and 1 stop bit.



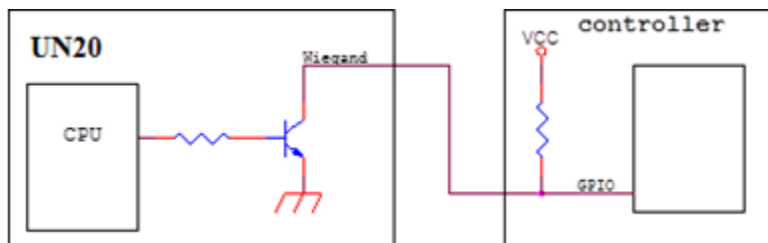
Communications Settings

If you connect UN20 Serial to a serial display, you will see not see any output upon boot up. This is because by default the **SDA20** fingerprint management system is listening on /dev/ttyS3.

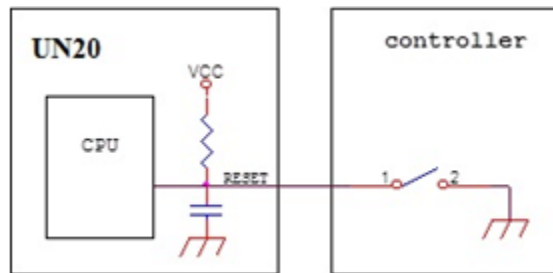
5.3. Relay



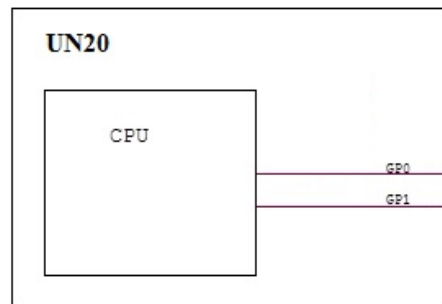
5.4. Wiegand



5.5. Reset



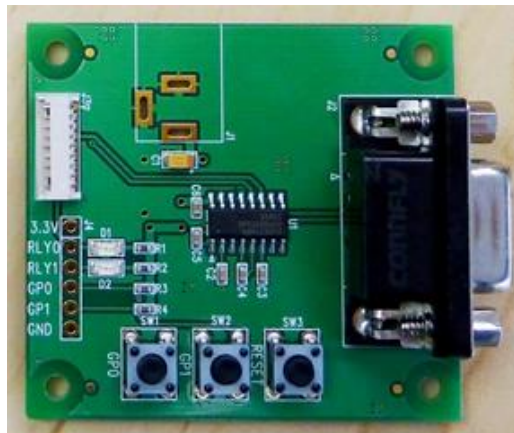
5.6. GPIO



GPIO ports are used as general purpose output only.

5.7. Using the DB9 Development Board

The UN20 Serial Developer Kit includes a development board that facilitates rapid integration and prototyping. This board connects to the UN20 Serial using the 9 pin Molex cable included with the kit. For general specifications, dimensions, and connector specifications, please refer to the **UN20 DK Board Datasheet**.



Appendix A. About SecuGen

About SecuGen

SecuGen (www.secugen.com) provides biometric products and development tools for development organizations that are creating physical and network security systems employing advanced fingerprint recognition technology. The company's comprehensive product line includes high quality optical fingerprint readers and sensor component, software and development kits that are used for developing innovative applications including Internet, enterprise network and desktop security, physical access control, time and attendance management and financial and medical records control. SecuGen patented products are renowned for their accuracy, reliability, ruggedness, and affordability. Based in Silicon Valley, SecuGen has been serving the global biometric community since 1998.

SecuGen Sensor Qualities

- **Excellent Image Quality:** Clear, distortion-free fingerprint images are generated using advanced, patented optical methods. Quality imaging yields better sampling for minutiae data extraction.
- **Durability:** Mechanical strength tests show resistance to impact, shock and scratches.
- **Powerful Software:** Precise, fast processing algorithm ensures efficiency and reliability.
- **Ruggedness and Versatility:** Solid engineering and superior materials allows for use under extreme conditions.
- **Ergonomic Design:** Compact, modular design for seamless integration into small devices, ease of use, and compatibility make SecuGen Sensors ideal for a broad range of applications.
- **Low Cost:** Products are developed to deliver high performance, with zero maintenance at very affordable prices for general and industrial use.

Advantages of SecuGen Sensors over Other Fingerprint Sensors

- Unique optical method captures fine details, even from dry skin
- Extremely low image-distortion for greater accuracy
- Resistance to damaging electrostatic discharge, moisture or corrosion
- Superior mechanical strength, wear-resistance and durability with no need for costly coatings
- Broad range of applicability, especially for use in extreme conditions and climates
- Low cost, long life, and no maintenance requirements – suitable for mass deployments

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